AI AUTOMATION & AGENTS

Build Workflows. Deploy Agents. Work Smarter.

Course 2 of the AI Education Series | Non-Credit Professional Development

University of Arizona Online

Course Information

Parameter	Details
Format	5 modules, self-paced online with optional instructor-led cohort sessions
Total Hours	40 hours (8 hours per module)
Level	Intermediate — requires completion of Course 1
Audience	Workforce professionals and graduate students seeking applied AI skills
Time Commitment	~8 hours per module
Tools Used	Claude Desktop (Cowork + Code), n8n, LangChain, LangGraph, CrewAI, Ollama, Google Colab
Assessment	Concept quizzes, guided lab exercises, one hands-on project per module, case study analyses, peer reviews, and a capstone deployment policy
Prerequisites	Basic Python knowledge
Free-Tier Path	A no-cost alternative track is available using Ollama, n8n Community Edition, and Google Colab

Course Description

This course moves beyond individual AI prompts into the world of AI agents and automated workflows — systems that can take actions, use tools, manage files, and complete multi-step tasks on your behalf. Learners will progress from no-code automation tools through to structured agent frameworks used in industry, gaining transferable skills applicable to any organization or platform.

The course is structured around a layered tool strategy: early modules introduce accessible no-code and low-code tools (Claude Cowork, n8n) to build automation intuition, while later modules introduce the leading open-source agent frameworks — LangChain, LangGraph, and CrewAI — through guided Jupyter Notebook labs on Google Colab. Throughout, the emphasis remains practical and workforce-relevant.

The course closes with a dedicated capstone module on responsible agentic AI, grounded in published frameworks including the NIST AI Risk Management Framework (AI RMF 1.0), covering how to deploy agents safely, audit their behavior, and communicate automation policies to institutional stakeholders.

Course Learning Outcomes

Upon successful completion of this course, learners will be able to:

- **Define and explain:** core AI agent concepts — the agent loop, tool use, memory, orchestration, and multi-agent architectures (Bloom's L1–L2)
- **Map and evaluate:** real workplace workflows to identify automation opportunities and select appropriate tools and frameworks (L2–L4)
- **Build and deploy:** no-code automation pipelines using Claude Cowork and n8n, connected to live external services (L3)
- **Configure and apply:** the Model Context Protocol (MCP) to integrate AI agents with filesystems, databases, and cloud APIs (L3)
- **Develop and run:** multi-agent systems using LangChain, LangGraph, and CrewAI in Python-based Jupyter Notebook environments (L3–L4)
- **Analyze and compare:** automation framework options — no-code, low-code, and code-first — and justify tool selection for a given workflow (L4)
- **Audit and document:** an AI workflow for safety, privacy, and reversibility, producing a responsible deployment policy aligned to the NIST AI RMF (L4)

Standard Activity Structure per Module (Bloom's Taxonomy)

Each module is structured to develop cognitive competencies across Bloom's Taxonomy Levels 1–4, from foundational recall to analytical reasoning. The following activity types are present in every module, yielding approximately 10 hours of instructional engagement per module.

Activity Type	Bloom's Level	Est. Hours	Description / Example
Reading + Video Content	L1 — Remember	1.5 h	Curated readings from official documentation, textbook chapters, or research summaries; 2–3 videos (15–25 min each) from high-reach channels (freeCodeCamp, LangChain, Fireship)
Concept Quiz / Retrieval Check	L1–L2	0.5 h	10–15 auto-graded multiple-choice and short-answer questions covering key vocabulary, framework concepts, and compare/contrast distinctions
Guided Lab Exercise	L2–L3 — Apply	2.0 h	Step-by-step tutorial using a provided dataset or scenario; replicates a demonstrated workflow before learners tackle the open-ended project
Hands-On Project	L3 — Apply	2.5 h	Open-ended applied project anchored to the learner's own workflow audit from Module 1; produces a deliverable submitted for grading
Case Study / Comparative Analysis	L4 — Analyze	1.0 h	Structured written analysis of a real-world scenario, framework comparison, or documented AI automation failure; uses a provided analytical rubric
Discussion / Peer Review	L2–L4	0.5 h	Asynchronous discussion prompt or structured peer review of a classmate's project deliverable or policy document

Bloom's Spiral Progression Across the Course

Modules 1–2: Weighted toward L1–L2 (Remember, Understand) — building vocabulary and mental models for automation design.

Modules 3–4: Weighted toward L2–L3 (Understand, Apply) — integrating tools and building functional agent systems.

Module 5: Weighted toward L3–L4 (Apply, Analyze) — auditing, policy-writing, and analytical evaluation of complete systems.

MODULE 1 **From Prompts to Pipelines: Thinking Like an Automation Designer**	**8 Hours**

Instructor Introduction

Every powerful automation begins not with a tool, but with a question: what is actually happening in this workflow, and where is the human effort most replaceable? Module 1 is intentionally tool-light. Before you touch Claude Cowork, n8n, or any agent framework, you need a mental model that will make every subsequent module coherent. This module builds that model.

We begin by defining what an AI agent is — precisely, not loosely — and distinguishing it from a chatbot, a script, or a search engine. We trace the agent loop (perceive → plan → act → observe) that underlies every automated system you will build in this course, from a simple Cowork file-organization task to a multi-agent LangGraph workflow. We then survey the full automation landscape — from no-code tools like n8n to orchestration frameworks like LangChain and CrewAI — so that you can see the whole map before we begin exploring individual territories.

The module's anchor activity is the Workflow Audit: a structured analysis of three repetitive tasks from your own work or studies. You will return to this audit in every subsequent module, progressively automating, connecting, and refining the workflows you identify here. Begin it thoughtfully.

Topics Covered

- What is an AI agent? Precise definitions: perception, planning, action, and observation
- The agent loop: perceive → plan → act → observe — illustrated with real examples
- Prompts vs. pipelines: why automation requires a different mindset than a single conversation
- The anatomy of a workflow: triggers, tasks, conditional logic, tools, and outputs
- Identifying automation candidates: high-repetition, rule-based, and time-consuming tasks
- Overview of the automation landscape: no-code (n8n, Cowork), low-code (LangChain), and code-first (custom Python agents)
- Introduction to key open-source frameworks: LangChain, LangGraph, CrewAI, Ollama — what each does and when to use it
- What agents can and cannot do: setting realistic expectations and avoiding common over-automation mistakes
- Introduction to the Claude Desktop app: Chat, Cowork, and Code tabs explained
- Key vocabulary: agent, pipeline, trigger, tool call, MCP, RAG, sandbox, orchestration, and 15+ additional terms

Learning Activities (Bloom's L1–L4)

Activity Type	Bloom's Level	Est. Hours	Description / Example
Reading + Video	L1 — Remember	1.5 h	Read: LangChain Introduction to Agents (docs); Watch: freeCodeCamp 'What are AI Agents?' (YouTube); Watch: Fireship 'AI Agents Explained' (YouTube)
Glossary Activity	L1 — Remember	0.5 h	Complete a 25-term illustrated glossary covering: agent, pipeline, trigger, tool call, MCP, RAG, orchestration, sandbox, vector database, multi-agent, LLM, context window, memory, embedding, etc.
Concept Quiz	L1–L2	0.5 h	15 auto-graded questions: definitions, agent loop sequencing, matching automation paradigms to use cases

Activity Type	Bloom's Level	Est. Hours	Description / Example
Guided Lab	L2–L3 — Apply	2.0 h	Structured walkthrough: (1) explore Claude Desktop tabs; (2) build and run a simple Ollama local model chat; (3) diagram a provided sample workflow using the worksheet template
Workflow Audit Project	L3 — Apply	2.5 h	Identify three workflows, complete the mapping worksheet for each (trigger, steps, tools, outputs), score on rule-based vs. high-stakes dimensions, and rank by automation potential
Comparative Analysis	L4 — Analyze	1.0 h	Written reflection: 'For one of your audited workflows, which automation paradigm — no-code, low-code, or code-first — is most appropriate? Compare at least two options using the criteria from the lecture.'

HANDS-ON PROJECT: Your Workflow Audit

Identify three repetitive tasks from your actual job or studies that currently take more than 30 minutes per week. For each task, complete the workflow mapping worksheet: document the trigger (what starts it), the steps involved, the tools or files touched, and the final output.

Assess each task on two dimensions: (1) how rule-based it is — can each step be precisely specified? — and (2) how high-stakes a mistake would be — what is the cost of an error? Rank all three tasks by automation potential.

Write a 200-word reflection on which automation paradigm (no-code, low-code, or code-first) you believe is most appropriate for your top-ranked task, and why. You will return to this audit in every subsequent module.

Curated Resources

Resource	URL / Source	Why It Matters
LangChain: Introduction to Agents	python.langchain.com/docs/concepts/agents	~95k GitHub stars; foundational reference for agent architecture concepts used throughout this course
What are AI Agents? (freeCodeCamp, YouTube)	youtube.com/c/freeCodeCamp	9M+ subscribers; accessible, rigorous introductions to AI concepts
Ollama — Run LLMs Locally (free)	ollama.com / github.com/ollama/ollama	~100k GitHub stars; enables free local model experimentation without a paid API subscription
n8n Official Documentation	docs.n8n.io	~50k GitHub stars; free, self-hostable workflow automation platform introduced in Module 2
Hugging Face: Models, Spaces & Datasets	huggingface.co	~40k+ stars (transformers); primary repository for pre-trained models, demo spaces, and public datasets

MODULE 2 **No-Code Automation: Claude Cowork, n8n & Connected Apps**	**8 Hours**

Instructor Introduction

Module 2 is where theory becomes practice. You will build your first real automated pipelines using two complementary no-code platforms: Claude Cowork, which is built into the Claude Desktop app and excels at language-centric tasks like document generation and research synthesis; and n8n, a free, open-source workflow automation platform with a visual canvas that lets you connect hundreds of applications without writing a line of code.

A key design choice in this module is the deliberate pairing of a proprietary tool (Cowork) with an open-source alternative (n8n). This is intentional: professional competence means being able to select and operate tools based on organizational requirements, cost constraints, and data governance policies — not allegiance to any single vendor. You will complete exercises in both tools and, in the case study activity, critically compare them on dimensions that matter in real institutional deployments.

By the end of Module 2 you will have built at least one end-to-end automated pipeline and produced your first Agent Instruction Log — a professional habit of documenting what instructions worked, what needed revision, and where the agent made errors. This log will become a portfolio artifact.

Topics Covered

- Getting started with Claude Cowork: granting folder access, giving tasks, reviewing and iterating on outputs
- What Cowork can do: research synthesis, file organization, document generation, formatted spreadsheets and presentations
- Giving effective Cowork instructions: specificity, scope, checkpoints, and iterative refinement
- Connecting Claude to external services: Google Drive, Gmail, Slack, Google Calendar via the Connectors menu
- Introduction to n8n: the visual canvas, nodes, triggers, and the community edition (free, self-hosted)
- Core automation logic: triggers, conditions, transformations, and actions — concepts applicable to any platform
- Building a 3-node n8n workflow: trigger → data transformation → output to a destination
- When Cowork is not the right tool: recognizing tasks that require structured logic, local execution, or data privacy
- Data governance primer: what information leaves your organization when you use cloud-based automation tools

NOTE: Platform Availability

Claude Cowork is available on paid Claude plans (Pro, Max, Team, Enterprise) via the Desktop app. It is not supported on Windows arm64 and is currently a research preview — features may change. All core concepts in this module are also demonstrated using n8n Community Edition, which is free and requires no subscription.

Learning Activities (Bloom's L1–L4)

Activity Type	Bloom's Level	Est. Hours	Description / Example
Reading + Video	L1 — Remember	1.5 h	Read: n8n Official Documentation — Getting Started; Watch: n8n official YouTube channel (workflow fundamentals); Read: Zapier University automation concepts (free, platform-agnostic)

Activity Type	Bloom's Level	Est. Hours	Description / Example
Concept Quiz	L1–L2	0.5 h	15 questions: automation terminology, trigger vs. action vs. condition, matching workflow types to use cases, identifying when cloud vs. self-hosted tools are appropriate
Guided Lab — n8n	L2–L3 — Apply	2.0 h	Step-by-step: install n8n Community Edition (or use n8n cloud trial); build a 3-node workflow: HTTP trigger → text transformation → save to Google Sheets or local file output
Hands-On Project	L3 — Apply	2.5 h	Build an end-to-end automated office productivity pipeline using Cowork (paid track) or n8n (free track) applied to one task from your Module 1 workflow audit
Comparative Case Study	L4 — Analyze	1.0 h	Written analysis: 'Does the workflow you built in Cowork expose any organizational data to external servers? How does that compare to a self-hosted n8n deployment? What are the organizational implications?' (400 words, structured rubric provided)
Peer Discussion	L2–L3	0.5 h	Asynchronous forum: post your Agent Instruction Log (what worked, what failed, what you changed) and respond to one classmate's log with a constructive observation

HANDS-ON PROJECT: Automated Office Productivity Pipeline

Choose one of the high-repetition tasks you identified in Module 1. Use Cowork or n8n to automate it end-to-end.

Example pipelines: (a) Cowork — grant access to a folder of meeting notes, instruct it to extract action items, assign owners, and produce a formatted follow-up email draft; (b) n8n — build a scheduled trigger that reads from a Google Sheet, filters rows by a condition, and writes a summary report to Google Drive.

Document your process in an Agent Instruction Log: record the instructions you gave, which worked on the first attempt, which needed revision, and what the agent got wrong. This log is a graded deliverable and a professional portfolio artifact.

Curated Resources

Resource	URL / Source	Why It Matters
n8n Official Documentation	docs.n8n.io	~50k GitHub stars; primary reference for no-code workflow automation; free community edition
n8n Official YouTube Channel	youtube.com/@n8n-io	Video tutorials for building workflows; practical, beginner-friendly
Zapier University (free)	learn.zapier.com	Platform-agnostic automation concepts; industry-recognized free certification
Claude Cowork Documentation	docs.claude.ai (search: Cowork)	Official reference for Cowork capabilities, connectors, and setup
Hugging Face Spaces	huggingface.co/spaces	Live demos of open-source AI tools; useful for exploring automation capabilities without installation

MODULE 3 **AI with Memory & Integrations: MCP, LangChain Tools & Agent Memory**	**8 Hours**

Instructor Introduction

Module 3 addresses one of the most important architectural questions in AI agent design: how does an agent connect to the world, and how does it remember what it has done? The answer to the first question is tool use, standardized through the Model Context Protocol (MCP). The answer to the second is memory — a concept that encompasses short-term context, long-term vector storage, and episodic records of tool calls and agent actions.

We approach MCP as an open standard — not merely as a feature of the Claude Desktop app. You will read the official MCP specification, understand its design goals, and compare it to LangChain's tool integration system, which is the most widely used agent-tool interface in the open-source ecosystem. This comparative perspective ensures that what you learn here transfers beyond any single product.

The Permission Log you began constructing during the Module 2 project grows significantly in this module. Every integration you build — filesystem, email, calendar, vector database — requires an explicit access decision, and every decision should be documented. This habit is both professionally valuable and a direct preparation for the Module 5 responsible deployment capstone.

Topics Covered

- What is the Model Context Protocol (MCP)? A plain-language explanation of the open standard for agent-tool communication
- MCP architecture: hosts, clients, servers, and the tool call lifecycle
- Desktop Extensions in the Claude app: browsing, installing, and configuring MCP servers
- Web connectors vs. desktop extensions: what's the difference and when to use each
- LangChain Tools and Toolkits: the parallel open-source standard for agent-tool integration used in production systems
- Building a multi-step integration: connecting calendar + email + a document store
- Agent memory architecture: three types
- Short-term memory: the context window — what fits, what doesn't, and why it matters
- Long-term memory: vector databases — embeddings, similarity search, and ChromaDB as a free open-source vector store
- Episodic memory: tool call logs — how agents record and replay prior actions
- Data flow awareness: understanding what information leaves your machine and where it goes
- Permission scoping: the principle of minimal authority in practice
- Troubleshooting integrations: when tools don't respond, return errors, or produce unexpected results

Learning Activities (Bloom's L1–L4)

Activity Type	Bloom's Level	Est. Hours	Description / Example
Reading + Video	L1 — Remember	1.5 h	Read: Official MCP Specification (modelcontextprotocol.io/docs); Read: LangChain Tools & Toolkits Guide (python.langchain.com/docs/concepts/tools); Watch: LangChain official 'Tools & Agents' YouTube explainer
Concept Quiz	L1–L2	0.5 h	15 questions: MCP architecture components, short/long-term/episodic memory distinctions, LangChain Tools vs. MCP comparison, data flow scenarios
Guided Lab — MCP + ChromaDB	L2–L3 — Apply	2.0 h	Step-by-step: (1) configure two MCP Desktop Extensions (e.g., filesystem + Google Drive); (2) run a Colab notebook

Activity Type	Bloom's Level	Est. Hours	Description / Example
			to create a ChromaDB collection, embed three documents, and run a similarity query — observing how an agent could use this for long-term memory
Hands-On Project	L3 — Apply	2.5 h	Build a Research Assistant Workflow connecting at least two external services; document every permission granted in your Permission Log
Memory Architecture Analysis	L4 — Analyze	1.0 h	Written case analysis: 'For your Module 1 top-ranked workflow, design an agent memory architecture. Which of the three memory types would you implement, and why? What would you store in the vector database, and what are the privacy implications?' (400 words + annotated diagram)
Peer Discussion	L2–L3	0.5 h	Forum: share your Permission Log and one unexpected data-flow finding with the group; respond to a classmate's log

HANDS-ON PROJECT: Research Assistant Workflow with Live Integrations

Build a Cowork-powered research and reporting pipeline that connects to at least two external services.

Example: connect Gmail and Google Drive, then instruct Claude to (1) scan a specified email label for research requests, (2) search a Drive folder for relevant existing documents, (3) synthesize a response incorporating both sources, and (4) save a formatted report back to Drive with a bibliography.

n8n alternative: build an equivalent workflow using n8n nodes for Gmail, Google Drive, and a text-output step — achieving the same integration without a paid Claude subscription.

Document every permission you grant and why, creating a Permission Log entry for each service. Note the data residency of each service (where your data goes). This log is a graded deliverable that feeds directly into the Module 5 capstone audit.

Curated Resources

Resource	URL / Source	Why It Matters
MCP Official Specification	modelcontextprotocol.io/docs	~14k GitHub stars; the open standard referenced throughout this module — required reading
LangChain Tools & Toolkits	python.langchain.com/docs/concepts/tools	~95k GitHub stars; most widely adopted agent-tool interface in production systems
ChromaDB Documentation	docs.trychroma.com	~15k GitHub stars; free, open-source vector store for agent long-term memory; runs locally or in Colab
Prefect Workflow Orchestration	docs.prefect.io	~18k GitHub stars; Python-native workflow orchestration; useful comparison to MCP-based designs
Hugging Face: Embedding Models	huggingface.co/models?pipeline_tag=feature-extraction	Free embedding models for populating vector

Resource	URL / Source	Why It Matters
		databases; no API key required

MODULE 4 **Building AI Agents: Claude Code, LangChain, LangGraph & CrewAI**	**8 Hours**

Instructor Introduction

This is the most technically demanding module in the course, and it is structured to be accessible to learners without a programming background. You will work in two parallel tracks: the Claude Code track, which lets you direct an AI that writes and executes code on your behalf through a visual interface; and the Open-Source Frameworks track, which introduces LangChain, LangGraph, and CrewAI through guided Jupyter Notebook labs on Google Colab — a free, browser-based Python environment requiring no local installation.

LangChain is the foundational framework for building agents that call tools and use LLMs programmatically. LangGraph extends LangChain with stateful, graph-based multi-agent workflows — the architecture behind most production-grade agentic systems. CrewAI offers a higher-level abstraction for defining role-based multi-agent teams (a Researcher agent, a Writer agent, a Critic agent) that collaborate on shared goals.

The module's capstone activity is a framework selection analysis: given the three workflows you audited in Module 1, which tool would you choose for each, and why? This is a genuine Level 4 analytical task — there is no single correct answer, and you will be evaluated on the quality of your reasoning, not its conclusion.

Topics Covered

Track A — Claude Code (Desktop App)

- Tour of the Code tab: sessions, local vs. remote, diff view, and permission modes
- Permission modes: Ask (default), Auto Accept, and Plan mode — when to use each
- Starting a session: selecting a project folder, giving Claude your first task
- Reviewing changes: reading diffs, accepting and rejecting edits, leaving comments for Claude
- Non-developer use cases: bulk file renaming, CSV-to-Excel conversion, automated report formatting, chart generation
- Using CLAUDE.md: giving Claude standing instructions about your project and preferences

Track B — Open-Source Agent Frameworks

- LangChain Agents and Tools: the ReAct pattern, tool definitions, and agent executors in Python
- Running LangChain agents with Ollama (free, local) as the LLM backend — no API costs
- Introduction to LangGraph: nodes, edges, state, and the StateGraph API
- Building a stateful multi-step agent graph with LangGraph: a research-and-summarize workflow
- Introduction to CrewAI: agents, tasks, roles, goals, and the Crew orchestration model
- CrewAI lab: defining a Researcher agent and a Writer agent working on a shared task
- Comparing frameworks: when to use LangChain, LangGraph, CrewAI, n8n, or Claude Code for a given workflow
- Multi-agent design principles: specialization, delegation, inter-agent communication, and error recovery

Learning Activities (Bloom's L1–L4)

Activity Type	Bloom's Level	Est. Hours	Description / Example
Reading + Video	L1 — Remember	1.5 h	Read: LangGraph 'Build Stateful Multi-Agent Apps' (langchain-ai.github.io/langgraph); Read: CrewAI Official Docs — Core Concepts; Watch: CrewAI 'Getting Started' official video; Watch: LangChain 'Agents' playlist

Activity Type	Bloom's Level	Est. Hours	Description / Example
Concept Quiz	L1–L2	0.5 h	15 questions: LangChain vs. LangGraph vs. CrewAI distinctions, agent loop components, ReAct pattern steps, multi-agent roles
Guided Lab A — Claude Code	L2–L3 — Apply	1.5 h	Provided messy dataset: use Claude Code in Plan mode to ingest, clean, and produce a formatted summary chart; save a CLAUDE.md with standing instructions for reuse
Guided Lab B — LangGraph Colab	L2–L3 — Apply	2.0 h	Google Colab notebook (provided): build a 3-node LangGraph StateGraph with Ollama as the LLM — perceive → plan → respond; inspect agent state at each step
Hands-On Project	L3 — Apply	2.5 h	CrewAI multi-agent pipeline: define a Researcher agent and a Writer agent, assign them a task from your Module 1 workflow audit, run the Crew, and review the collaborative output
Framework Selection Analysis	L4 — Analyze	1.0 h	For each of your three audited workflows from Module 1, identify the most appropriate framework (n8n, Claude Code, LangChain, LangGraph, or CrewAI). Write a 150-word justification for each choice using the comparison criteria from lecture. This is a graded Level 4 deliverable.

HANDS-ON PROJECT: Data Processing & Multi-Agent Pipeline

Track A (Claude Code): Use Claude Code in the Desktop app to build an automated data pipeline. Complete at least three steps: (1) ingest and clean a messy data file, (2) perform a basic analysis or transformation, (3) produce a formatted output (chart, summary table, or report). Use Plan mode first. Save your CLAUDE.md so the pipeline is reusable.

Track B (Open-Source): Using the provided Google Colab notebooks, build a CrewAI Crew with at least two specialized agents (e.g., Researcher + Writer) that collaborate on a task from your workflow audit. Use Ollama as the LLM backend (free). Document the agent roles, goals, and the final output produced.

All learners complete the Framework Selection Analysis as a standalone graded deliverable.

Curated Resources

Resource	URL / Source	Why It Matters
LangGraph: Build Stateful Multi-Agent Apps	langchain-ai.github.io/langgraph	~14k GitHub stars; official docs for stateful agent graph construction
CrewAI Official Documentation	docs.crewai.com	~27k GitHub stars; role-based multi-agent orchestration; beginner-friendly design
LangChain Python Documentation	python.langchain.com/docs	~95k GitHub stars; foundational agent framework reference
Ollama — Run LLMs Locally	ollama.com / github.com/ollama/ollama	~100k GitHub stars; free local LLM backend; eliminates API costs for all open-source labs
OpenAI Agents SDK	github.com/openai/openai-agents-python	~8k GitHub stars; useful comparison to Claude Code

Resource	URL / Source	Why It Matters
		for structured agentic task design
Google Colab (Free Python Notebooks)	colab.research.google.com	Free browser-based Python environment; no installation required; GPU access available

MODULE 5 **Responsible Agentic AI: Trust, Control & Safe Deployment**	**8 Hours**

Instructor Introduction

Agents that can autonomously read files, send emails, run code, and browse the web are genuinely powerful. That power comes with commensurate responsibility. Module 5 is the capstone of this course, and its question is not "what can agents do?" — you have answered that across four modules — but "what should agents be allowed to do, under what conditions, with what oversight, and documented how?"

This module is grounded in published responsible AI frameworks that carry institutional and regulatory weight: the NIST AI Risk Management Framework (AI RMF 1.0), the Anthropic Model Specification, and the EU AI Act risk category taxonomy. These are the same documents referenced in enterprise AI governance reviews, IRB protocols, and organizational AI use policies — and your ability to engage with them fluently is a professional differentiator.

The capstone deliverable — a Responsible Deployment Policy for one of your course workflows — is a document you can include in a professional portfolio and present to an employer, IT department, or institutional review board. It synthesizes everything you have learned: the workflow you audited in Module 1, the permissions you logged in Module 3, the framework you selected in Module 4, and the safety principles you encounter here.

Topics Covered

- The agentic risk spectrum: low-stakes automation (file renaming) vs. high-stakes autonomous action (sending emails, executing financial transactions)
- What are agentic sandboxes? How Cowork's isolated VM model limits blast radius — and how to apply the same principle to any framework
- Permission models in practice: filesystem scoping, read-only vs. read-write, network access controls
- The principle of minimal authority: giving agents only the access they need — applied across Claude, n8n, LangChain, and CrewAI
- Human-in-the-loop design: where to require approval, where full automation is safe, and how to design approval checkpoints into agent workflows
- What can go wrong: a taxonomy of agentic AI failures — hallucination at scale, cascading tool errors, prompt injection, data exfiltration
- Prompt injection and adversarial inputs: what they are, how they work in agentic systems, and how to recognize them
- Privacy and data governance: what agent workflows should never touch — PII, credentials, regulated data, privileged communications
- Responsible AI frameworks: NIST AI RMF 1.0 (governance, map, measure, manage), EU AI Act risk categories, Anthropic's responsible scaling policy
- Institutional policies: how to present an AI automation proposal to your employer, IT department, or institutional review board
- Accountability: why "the AI did it" is never a complete answer — individual and organizational responsibility in agentic deployments

Key Concepts: The Five-Dimension Audit Framework

Audit Dimension	Key Question	Typical Findings
1. Data Sensitivity	What information does the agent access or transmit?	PII exposure, credential handling, regulated data (FERPA, HIPAA)
2. Reversibility	Can mistakes be undone? What is the blast radius of an error?	Irreversible: sent emails, deleted files, published posts. Reversible: draft documents, local reports

Audit Dimension	Key Question	Typical Findings
3. Permissions	Does the agent have more access than it needs?	Over-scoped filesystem access, unnecessary write permissions, unused connector authorizations
4. Human Checkpoints	Where does a human review or approve outputs?	Missing approval gates before external actions; no logging of agent decisions
5. Failure Modes	What happens if the agent produces incorrect results?	No fallback behavior, no error notification, no graceful degradation

Learning Activities (Bloom's L1–L4)

Activity Type	Bloom's Level	Est. Hours	Description / Example
Reading + Video	L1 — Remember	1.5 h	Read: NIST AI Risk Management Framework (AI RMF 1.0) — Executive Summary and Core sections; Read: Anthropic Model Specification (anthropic.com/model-specification) — selected sections; Watch: NIST AI RMF overview webinar (nist.gov)
Concept Quiz	L1–L2	0.5 h	15 questions: NIST AI RMF core functions, risk category definitions, minimal authority principle, prompt injection recognition scenarios
Case Study Analysis	L4 — Analyze	1.0 h	Analyze a provided documented agentic AI failure (real published incident): identify which of the five audit dimensions were violated, which safeguards were absent, and what specific change would have prevented the failure. Write a 500-word structured analysis using the Five-Dimension Audit rubric.
Workflow Audit	L3 — Apply	2.5 h	Apply the Five-Dimension Audit to one workflow you built in Modules 2, 3, or 4, using the provided audit rubric. Score each dimension and identify the highest-priority remediation action.
Responsible Deployment Policy	L3–L4 — Apply/Analyze	2.0 h	Write a one-to-two-page Responsible Deployment Policy for your audited workflow, using the provided template aligned to the NIST AI RMF. Include: workflow description, data sensitivity classification, permission inventory, human checkpoint design, failure mode response plan, and institutional approval recommendation.
Peer Review	L2–L4	0.5 h	Asynchronous peer review: review one classmate's Deployment Policy using the structured rubric; provide 200 words of specific, constructive feedback on one strength and one gap.

HANDS-ON PROJECT: Workflow Audit & Responsible Deployment Policy (Capstone)

Select one of the workflows you built in Modules 2, 3, or 4. Using the provided audit rubric, evaluate it across all five dimensions: (1) data sensitivity, (2) reversibility, (3) permissions, (4) human checkpoints, and (5) failure modes.

Then write a Responsible Deployment Policy for this workflow, using the template aligned to the NIST AI Risk Management Framework. The policy should be 1–2 pages and suitable for sharing with a manager, IT department, or institutional review board.

Submit both the completed audit rubric and the policy document. Participate in the asynchronous peer review by reviewing one classmate's policy and providing structured written feedback using the peer review rubric.

This deliverable is the course capstone and the most significant portfolio artifact you will produce. It demonstrates professional readiness to deploy AI automation responsibly in organizational settings.

Curated Resources

Resource	URL / Source	Why It Matters
NIST AI Risk Management Framework (AI RMF 1.0)	nist.gov/system/files/documents/2023/01/26/AI RMF 1.0.pdf	U.S. Federal standard for AI risk governance; widely cited in enterprise and academic AI policy
Anthropic Model Specification	anthropic.com/model-specification	Anthropic's published guidelines on responsible AI behavior; primary industry reference for this course
EU AI Act: Risk Category Overview	artificialintelligenceact.eu	Accessible summary of EU AI Act risk tiers; relevant for internationally-facing organizations
OWASP Top 10 for LLM Applications	owasp.org/www-project-top-10-for-large-language-model-applications	Authoritative security reference for prompt injection, data leakage, and other LLM-specific risks
AI Incident Database	incidentdatabase.ai	Curated repository of documented AI failure incidents; primary source for the Module 5 case study

Pre-Course Setup Checklist

Complete these steps before Module 1. A free-tier alternative path is provided for learners who do not have a paid Claude subscription. If you encounter issues, bring them to the first session — setup time will be available.

✓	Item	Paid-Plan Path	Free Alternative
☐	Claude Account	Pro, Max, Team, or Enterprise — upgrade at claude.ai/upgrade	Free Claude.ai account sufficient for Modules 1–2 conceptual content
☐	Claude Desktop App	Download from claude.ai/download (macOS + Windows x64)	Not required for free-tier path; conceptual exercises use claude.ai browser
☐	Verify App Tabs	Open app; confirm Chat, Cowork, and Code tabs visible at top	N/A
☐	Install Code Dependencies	Click Code tab; follow one-time prompt to install runtime dependencies	N/A
☐	Git (Windows only)	Install from git-scm.com/install/win; accept all defaults	Required for both paths if on Windows
☐	Enable Connectors	Settings > Connectors: connect Google Drive and Gmail	Connect same services via n8n Community Edition (free)
☐	Ollama (all learners)	Install from ollama.com; run: ollama pull llama3.2	Primary free LLM backend for Modules 3–4 open-source labs
☐	n8n Community Edition	Install from docs.n8n.io/hosting (self-hosted) or start n8n cloud free trial	Primary no-code automation platform for Module 2 free track
☐	Google Colab Access	Verify access at colab.research.google.com (requires Google account)	Required for all learners — Modules 3 and 4 labs run in Colab
☐	Practice Folder	Create a folder on your desktop called 'Claude Projects'	Same — create 'AI Automation Projects' folder for n8n outputs

Course Map

Module	Title	Primary Tools	Hours	Capstone Deliverable
M1	From Prompts to Pipelines	Claude Desktop, Ollama	8 h	Workflow Audit + Comparative Reflection
M2	No-Code Automation	Cowork + n8n + Connectors	8 h	Office Pipeline + Agent Instruction Log
M3	MCP, Tools & Agent Memory	MCP + LangChain Tools + ChromaDB	8 h	Research Assistant Workflow + Permission Log
M4	Building AI Agents	Claude Code + LangGraph + CrewAI	8 h	Data Pipeline + Framework Selection Analysis
M5	Responsible Agentic AI (Capstone)	NIST AI RMF + Audit Rubric	8 h	Workflow Audit + Deployment Policy + Peer Review

Completion & Certification

- Complete all five module projects and pass all five concept quizzes (70% or above) to earn a digital certificate of completion.
- Learners who complete both Course 1 (Introduction to Generative AI) and Course 2 (AI Automation & Agents) will receive the University AI Education Series professional badge.
- A free-tier completion path is supported: learners who complete all activities using Ollama, n8n, and Google Colab in lieu of paid Claude tools are eligible for the same certificate.
- Bring your own work: learners are strongly encouraged to apply course projects to real tasks from their jobs or studies. The Workflow Audit, Permission Log, and Deployment Policy are portfolio-quality artifacts.
- This syllabus is subject to change as AI tools evolve. Course materials will be updated to reflect significant changes to tools, frameworks, or published standards. Check the course portal for the latest version.

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